Example 1.18 If the function $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=\begin{cases}2x+7; & x<-2\\ x^2-2; & -2\le x<3\\ 3x-2; & x\ge 3\end{cases}$, then find the values of

(i) $f(4)$ (ii) $f(-2)$ (iii) $f(4)+2f(1)$ (iv) $\dfrac{f(1)-3f(4)}{f(-3)}$

Solution

The function f is defined by three values in intervals I, II, III as shown by the side

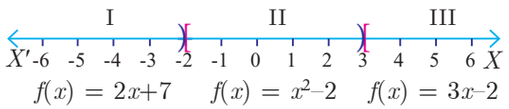


Fig. 1.39

For a given value of $x=a$, find out the interval at which the point a is located, there after find $f(a)$ using the particular value defined in that interval.

(i) First, we see that, $x=4$ lie in the third interval.

$\therefore\ f(x)=3x-2;\ f(4)=3(4)-2=10$

(ii) $x=-2$ lies in the second interval.

$\therefore\ f(x)=x^2-2;\ f(-2)=(-2)^2-2=2$

(iii) From (i), $f(4)=10$.

To find $f(1)$, first we see that $x=1$ lies in the second interval.

$\therefore\ f(x)=x^2-2\Rightarrow f(1)=1^2-2=-1$

$$f(4)+2f(1)=10+2(-1)=8$$

(iv) We know that $f(1)=-1$ and $f(4)=10$.

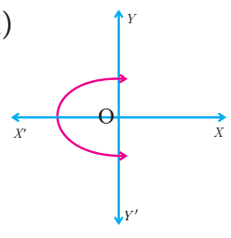
For finding $f(-3)$, we see that $x=-3$, lies in the first interval.

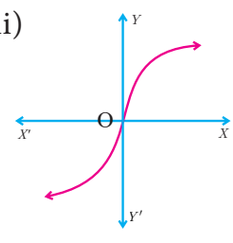
$\therefore\ f(x)=2x+7$; thus, $f(-3)=2(-3)+7=1$

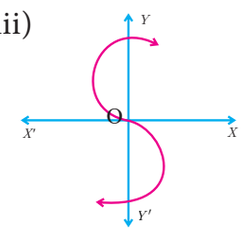
Hence, $\dfrac{f(1)-3f(4)}{f(-3)}=\dfrac{-1-3(10)}{1}=-31$

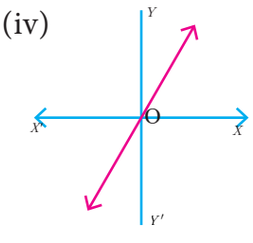
Exercise 1.4

1. Determine whether the graph given below represent functions. Give reason for your answers concerning each graph.

(i) 

(ii) 

(iii) 

(iv) 

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